

FUNCTIONAL TEST SUITE: REPORTS SECTION

Application: Business Management System | Feature: Reports Generation, Data Export & Access C

TC ID	Priority	Test Category	Description
TC_REP_001	P0	Positive / Export PDF	Verify successful generation and export of Sales Summary Report in PDF format with valid date range
TC_REP_002	P0	Positive / Export Excel	Verify export of Financial Ledger Report to Excel (.xlsx) with correct data schema, column types, and formula calculations
TC_REP_003	P0	Positive / Export CSV	Verify export of Customer Inventory Report in CSV format and validate UTF-8 encoding / comma delimiter structure
TC_REP_004	P0	Negative / Validation	Verify system error handling when attempting to generate a report with an invalid date range (End Date prior to Start Date)
TC_REP_005	P1	Boundary / Empty Data	Verify system behavior and report export when filtered criteria yield zero matching records

TC_REP_006	P1	Boundary / Large Dataset	Verify system stability, pagination/streaming, and memory usage when generating a large report (>100,000 records)
TC_REP_007	P1	Negative / Mandatory Filter	Verify report generation fails gracefully when mandatory filter parameters are not selected
TC_REP_008	P0	Role-Based Access Control	Verify restricted users ('Standard Employee') cannot access or generate Confidential Executive Payroll Reports
TC_REP_009	P1	Data Integrity Validation	Verify exported PDF/Excel report totals match screen preview summary metrics down to 2 decimal places
TC_REP_0010	P2	Concurrent Export Requests	Verify system stability and queue handling when multiple report export requests are initiated concurrently by the same user

Preconditions	Test Steps	Test Data
1. User logged in as Admin/Manager. 2. Sales transaction data exists in DB for Q3 2026.	1. Navigate to Reports > Sales Summary. 2. Select Date Range: '2026-07-01 to 2026-09-15'. 3. Select Export Format: 'PDF'. 4. Click 'Generate & Export Report'. 5. Open downloaded PDF file.	Report Type: Sales Summary Date Range: 01/07/2026 - 15/09/2026 Format: PDF
1. Active manager session. 2. Financial entries exist for selected department.	1. Navigate to Reports > Financial Ledger. 2. Set Filter Department: 'Operations'. 3. Select Format: 'Excel (.xlsx)'. 4. Click 'Export'. 5. Open exported Excel workbook.	Report Type: Financial Ledger Department: Operations Format: Excel (.xlsx)
1. Customer inventory data with special characters (UTF-8) exists in DB.	1. Navigate to Reports > Inventory Report. 2. Select Format: 'CSV'. 3. Click 'Download CSV'. 4. Inspect raw CSV via text editor and standard spreadsheet viewer.	Report Type: Inventory Report Format: CSV Encoding: UTF-8
1. Reports page open.	1. Navigate to Reports > Revenue Report. 2. Enter Start Date: '2026-09-01'. 3. Enter End Date: '2026-08-01' (prior to start date). 4. Click 'Generate Report'.	Start Date: 2026-09-01 End Date: 2026-08-01 Format: PDF
1. Target date range has no transactions recorded.	1. Navigate to Reports > Audit Logs. 2. Select Date Range: '2010-01-01 to 2010-01-02' (no data period). 3. Select Format: 'Excel'. 4. Click 'Export'.	Date Range: 2010-01-01 to 2010-01-02 Format: Excel

1. DB seeded with 150,000 transaction records for FY 2025-2026.	1. Navigate to Reports > Detailed Transactions. 2. Select Date Range: '2025-01-01 to 2026-09-01'. 3. Select Format: 'CSV'. 4. Initiate export and monitor server/browser response.	Record Count: ~150,000 rows Format: CSV
1. Reports page active.	1. Navigate to Reports > Expense Analysis. 2. Leave mandatory 'Business Unit' dropdown unselected. 3. Click 'Generate Report'.	Mandatory Field: Business Unit (Left Blank)
1. User logged in with 'Standard Employee' role (lacks 'ROLE_EXECUTIVE_REPORTS').	1. Attempt to navigate directly to '/reports/executive-payroll'. 2. Attempt API request 'GET /api/v1/reports/payroll?format=pdf' via HTTP client.	User Role: Standard Employee Target: Executive Payroll Report
1. Report preview generated on UI dashboard.	1. Navigate to Reports > Tax Summary. 2. Generate UI preview and note total tax calculated: '\$14,520.85'. 3. Click 'Export to Excel'. 4. Compare UI summary metrics with exported file cell values.	UI Preview Total: \$14,520.85 Report Format: Excel / PDF
1. User logged into application on multi-tab setup.	1. Open 3 browser tabs pointing to heavy reports (Sales, Audit, Inventory). 2. Click 'Export PDF' simultaneously across all three tabs within 2 seconds.	Concurrent Jobs: 3 simultaneous heavy PDF exports

Expected Result

1. PDF downloads successfully with proper filename ('Sales_Summary_20260701_20260915.pdf').
2. Data matches DB records accurately.
3. Header, column alignments, page numbers, and total summaries display correctly without overlapping.

1. Excel file downloads without corruption warnings.
2. Column headers match UI labels.
3. Numeric/currency values are formatted as numbers/currency (not text).
4. Total row contains functional SUM formulas ('=SUM(...)').

1. CSV downloads correctly.
2. Fields containing commas or quotes are properly escaped with double quotes.
3. Special characters (e.g., non-English names) render properly without encoding artifacts.

1. Report generation is blocked.
2. Inline validation error displays: 'End Date cannot be earlier than Start Date.'
3. Export buttons remain disabled.

1. UI displays notification: 'No records found for the selected criteria.'
2. If empty export is permitted, generated file contains standard headers with 0 data rows and 'No Data Available' summary row.

1. Application processes request using streaming/asynchronous download without timing out or throwing HTTP 504/500.
2. Complete dataset (150,000+ rows) exported accurately.

1. Field validation highlights 'Business Unit' in red.
2. Error text displayed: 'Please select a Business Unit to generate this report.'
3. No backend report job is spawned.

1. Navigation redirected to Unauthorized / 403 Access Denied page.
2. Direct API request returns HTTP 403 Forbidden with payload `{"message": "Access Denied"}`.

1. Exported document totals match UI preview exactly (\$14,520.85).
2. No floating point rounding errors or truncated decimals observed.

1. System queues or processes export requests gracefully.
2. All three reports complete downloading successfully without throwing HTTP 500 or corrupting file outputs.